

TLP:CLEAR

Honeypot Threat Intelligence

Monthly Report — April 2026

SSH Brute-force | WannaCry/EternalBlue | LLM Abuse | ICS/OT | SIP Scanning

Parameter	Value
Report period	April 1-30, 2026
Analysis date	May 7, 2026
Classification	TLP:CLEAR
Threat level	MEDIUM-HIGH
Honeypot nodes	DE (Hetzner), US (HOSTKEY), RU (SelectEI), PL (new — ICS/OT)
SSH events	578,657 login attempts across all nodes
Malware samples	654 unique (992 scanned, 100% malicious)
Abuse reports	184 generated, 35 submitted to ISPs

1. Executive Summary

During April 2026, the distributed honeypot network — spanning nodes in Germany (DE), the United States (US), and Russia (RU), with a new ICS/OT node launched in Poland (PL) on April 28 — recorded sustained and diverse attack activity across all monitored services. The infrastructure collectively observed 578,657 SSH login attempts, captured 654 unique malware samples (all confirmed malicious), and logged more than 236,000 SIP/VoIP scanning events. These figures reflect a threat landscape that remains active and operationally diverse, with attackers pursuing credential access, cryptomining, botnet expansion, and increasingly, abuse of AI inference infrastructure.

Several findings are of elevated concern. WannaCry-matching PE32 binaries continue to be distributed at scale — 884 samples triggered the WannaCry_Ransomware YARA rule — underscoring that legacy ransomware propagation mechanisms remain a live threat even in 2026. A coordinated SSH brute-force campaign specifically targeted Solana validator usernames (sol, solana, solv), with over 37,000 combined attempts, indicating deliberate targeting of blockchain infrastructure. Separately, 54,032 unauthorized inference requests were logged against the Ollama API honeypot, while Docker API events included 1,049 sessions classified as LLM-abuse — reflecting a growing attacker interest in hijacking exposed AI services for compute resource theft.

Infrastructure attribution reveals heavy reliance on commercial cloud providers. DigitalOcean accounts for the largest single source of correlated attacker traffic (1,835 hits), followed by OVH and Amazon AWS. Notably, 21 attack-correlated events were traced to Aeza International, a hosting provider subject to European and US sanctions, and 8 to Proton66 OOO, a known bulletproof hosting operator. TLS traffic analysis identified 3,383 sessions terminating at api[.]telegram[.]org, consistent with malware families using Telegram Bot API as a command-and-control (C2) channel. A Go-based SSH worm matching the Panchan signature was observed, with samples using Discord as a secondary C2 mechanism. The new Polish ICS/OT honeypot, emulating Modbus, IEC 61850, DNP3, and S7 protocols, began attracting traffic within hours of launch; full statistical analysis will follow in the May report.

2. Infrastructure Overview

The honeypot network consists of four nodes deployed across geographically and jurisdictionally diverse hosting environments. Each node runs a standardized suite of emulated services. The US node mirrors the full HOSTKEY data center traffic, meaning its raw connection counts include transit and neighboring-tenant traffic in addition to direct attacks on the honeypot; attack-specific analysis uses filtered datasets.

Node	Location / Provider	Protocols Monitored	Status
DE	Hetzner, Falkenstein, Germany	SSH (Cowrie), HTTP/HTTPS, SMB (:445), Ollama API (:11434), Docker API, WinBox, Kubelet, SIP/VoIP	Active — full month

US	HOSTKEY, New York, USA	SSH (Cowrie), HTTP/HTTPS, SIP/VoIP, Docker API (full DC mirror)	Active — full month; transit-filtered for attack stats
RU	SelectEI, Moscow, Russia	SSH (Cowrie), SIP/VoIP, HTTP/HTTPS	Active — full month
PL	New node, Poland	Modbus, IEC 61850, DNP3, S7 (ICS/OT emulation)	Launched April 28 — 3 days data only

3. Attack Volume & Trends

SSH login attempt volumes per node for April 2026. The DE node received the highest absolute volume of failed attempts, while the RU node shows a notably lower success-to-total ratio. Across all nodes, 30,984 login attempts were recorded as "successful" by the honeypot emulation layer, providing visibility into post-exploitation commands and payloads.

Node	Successful Logins	Failed Logins	Total Attempts
DE	14,886	237,636	252,522
US	13,232	139,888	153,120
RU	2,866	170,149	173,015
Total	30,984	547,673	578,657

Honeypot event volumes by service type for April 2026:

Service / Honeypot	Events / Sessions	Notes
SSH (Cowrie)	578,657 login attempts	35 active campaigns; 1,921 unique attacker IPs
Ollama API	54,032 events	Unauthorized LLM inference attempts
SIP/VoIP	236,573 total events	DE: 46,316 RU: 79,342 US: 110,915
Docker API	7,834 events	Breakdown by category in section 4.3
WinBox (MikroTik)	7,442 events	MikroTik admin interface scanning
Kubelet (Kubernetes)	1,494 events	Kubernetes API server scanning

4. Key Findings

4.1 WannaCry/EternalBlue Still Distributing via SMB

The most prevalent YARA match across collected malware was WannaCry_Ransomware, triggering on 884 PE32 samples — the single largest homogeneous detection cluster of the month. A correlated rule, SUSP_Imphash_Mar23_2, matched the same 884 samples, consistent with the well-documented import-hash fingerprint shared across WannaCry variants. These samples were collected primarily via the dionaea honeypot on port 445 (SMB), which received 80,500 connection attempts on the DE node alone.

The continued propagation of WannaCry-derived binaries nearly a decade after the original 2017 outbreak reflects two persistent realities: a large population of unpatched Windows systems exposed on the internet that remain susceptible to EternalBlue (CVE-2017-0144 / MS17-010), and the ongoing recycling of the leaked NSA exploit toolchain within commodity malware. While the ransomware payload in many observed samples may be non-functional due to sinkholed killswitch domains, the underlying worm propagation mechanism is fully operational.

4.2 Solana Validator Brute-Force Campaign

A statistically significant cluster of SSH brute-force attempts targeted usernames associated with Solana blockchain validator infrastructure. Across all honeypot nodes, the usernames sol (18,399 attempts), solana (11,509 attempts), and solv (7,063 attempts) collectively accounted for 37,021 login attempts — placing them at positions 4, 6, and 7 respectively in the global top-15 username list, immediately behind root (226,565), admin (28,348), and ubuntu (24,957).

The concentration and specificity of these usernames indicates a deliberate campaign targeting Solana validator nodes rather than opportunistic credential stuffing. Solana validators are economically incentivized targets: a compromised validator can be used to extract staking keys, redirect validator rewards, disrupt consensus participation, or serve as a pivot into associated hot-wallet infrastructure. The presence of the username solv — associated with the Solv Protocol staking product — suggests attackers are tracking ecosystem-specific tooling and adjusting their credential lists accordingly.

4.3 LLM Infrastructure Abuse (Ollama + Docker API)

Two honeypot services recorded significant abuse of AI/LLM infrastructure. The Ollama API honeypot (emulating an unauthenticated Ollama inference endpoint on port 11434) logged 54,032 events during April, representing unauthorized attempts to use the service for text generation inference — effectively hijacking compute resources to run language model queries at no cost to the attacker. The DE node's port 11434 received 95,146 Zeek connection-level events, confirming sustained interest in this attack surface.

The Docker API honeypot logged 7,834 total events. Of these, 1,049 sessions were classified under the llm-abuse category, indicating attempts to deploy LLM inference containers via exposed Docker APIs. The Docker API breakdown for April 2026:

Session Category	Count	Description
recon	1,338	API enumeration and capability discovery
llm-abuse	1,049	Attempts to deploy or invoke LLM inference containers
unknown	507	Unclassified sessions
docker-worm	481	Automated worm propagation via Docker API
mikrotik-recon	38	MikroTik RouterOS detection via Docker API
malware-dropper	30	Malware payload delivery via container execution
cryptominer	2	Cryptocurrency mining container deployment

4.4 Telegram Bot API as Malware C2 Channel

Zeek SSL log analysis identified 3,383 TLS sessions terminating at the SNI `api[.]telegram[.]org` across the DE and RU nodes. This traffic did not originate from the honeypot operators — no Telegram-related tooling is deployed on honeypot infrastructure — and is therefore attributable to malware executing within the honeypot environment using the Telegram Bot API as a command-and-control channel.

The use of Telegram as a C2 transport is well-established across multiple malware families. Telegram's Bot API is attractive to malware authors because it is encrypted (TLS to Telegram's infrastructure), does not require the attacker to operate dedicated C2 infrastructure, is difficult to block without impacting legitimate Telegram traffic, and bot tokens provide a low-cost, disposable authentication mechanism. The volume of 3,383 sessions over one month indicates multiple distinct malware samples or persistent activity from recurring infections.

4.5 Panchan SSH Worm with Discord C2

Seven ELF samples triggered the `Discord_C2_ELF` YARA rule, and six samples matched the `Panchan_SSHWorm_Go` signature. All seven `Discord_C2_ELF` samples also triggered `SUSP_ELF_LNX_UPX_Compressed_File`, indicating UPX packing consistent with Panchan's known packaging methodology. Panchan is a peer-to-peer Go-based SSH worm first documented in 2022; it spreads via credential brute-force and SSH key theft, then establishes a mesh C2 network augmented by Discord webhook communications.

The MIPS and ARM ELF samples observed in April's collection (14 ARM, 4 MIPS across all ELF samples) are consistent with Panchan's cross-architecture targeting of IoT and embedded Linux devices alongside standard x86-64 servers. Discord webhooks provide attackers with a real-time notification channel for newly compromised hosts without requiring any dedicated infrastructure.

4.6 Mass SIP/VoIP Scanning

SIP/VoIP scanning activity was among the highest-volume observations of the month, with 236,573 total events across all three primary nodes: DE (46,316 Zeek SIP events), RU (79,342

events, including 34,863 Zeek SIP log entries), and US (110,915 events). The RU node's port 5060 received 65,815 connection-level events, and the DE node recorded 46,316 on the same port, confirming SIP as a primary scan target.

SIP scanning is typically conducted for two purposes: discovery of exposed VoIP infrastructure for toll fraud (unauthorized call placement through hijacked SIP accounts or PBX systems), and fingerprinting of VoIP endpoints for subsequent exploitation. The scale of scanning observed — over 230,000 events in a single month — reflects highly automated, internet-wide SIP enumeration campaigns.

4.7 Attacker Infrastructure: Sanctioned and Bulletproof Hosting

Correlator analysis of attacker ASN attribution identified attack-correlated traffic from two notable hosting providers. Aeza International generated 21 correlated hits; it is a hosting provider subject to sanctions by the European Union and the United States in connection with its role in providing infrastructure used for cybercriminal activity. Proton66 OOO generated 8 correlated hits; it is a Russian autonomous system widely documented as a bulletproof hosting provider used by ransomware operators, initial access brokers, and other cybercriminal groups.

The dominant attacker-infrastructure provider by volume was DigitalOcean (1,835 correlated hits), followed by OVH (436) and Amazon AWS (430). The concentration of attacker traffic in major commercial cloud providers reflects the operational preference for using low-cost, easily provisioned VPS infrastructure that can be rapidly replaced when blocked or reported.

5. Malware Analysis

A total of 654 unique malware samples were indexed during April 2026, collected from three sources: dionaea (578 samples, primarily PE32 via SMB/HTTP), cowrie (57 samples, dropped by SSH session attackers), and a URL harvester tracking dropper URLs (19 samples). All 992 samples submitted to the malware scanner received a malicious verdict. By primary node: DE contributed 298 samples, US 228, and RU 128.

File type distribution:

File Type	Count	Share
PE32/DLL (Windows)	930	93.8%
ELF (Linux)	51	5.1%
Shell scripts	7	0.7%
Other	4	0.4%

Linux ELF samples by CPU architecture:

Architecture	Sample Count
x86-64	25

ARM	14
MIPS	4
i386	3
ARC	1

Top YARA rule hits for April 2026:

YARA Rule	Hits	Significance
WannaCry_Ransomware	884	WannaCry-family PE32; SMB propagation via EternalBlue
SUSP_Imphash_Mar23_2	884	Correlated imphash fingerprint — same sample cluster
Go_SSHWorm_Generic	7	Generic Go SSH worm detection
Discord_C2_ELF	7	ELF with Discord webhook C2 capability
SUSP_ELF_LNX_UPX_Compressed_File	7	UPX-packed ELF — evasion packing
Panchan_SSHWorm_Go	6	Panchan P2P Go SSH worm
SUSP_XORed_Mozilla_Oct19	3	XOR-obfuscated strings (Mozilla/Firefox pattern)
MAL_ARM_LNX_Mirai_Mar13_2022	2	Mirai botnet — ARM Linux variant
MAL_ELF_Xlogin_Nov24_1	1	Xlogin ELF credential stealer

C2 indicators extracted from malware samples:

Indicator	Type	Associated Malware / Purpose
xmrig[.]com	Domain (defanged)	XMRIg cryptominer — mining client distribution
nicehash[.]com	Domain (defanged)	NiceHash mining pool — cryptomining payout
api[.]telegram[.]org	Domain (defanged)	Telegram Bot API — malware C2 channel
discord[.]com / discordapp[.]com	Domain (defanged)	Discord webhook — Panchan/Discord_C2_ELF C2
api[.]ipify[.]org	Domain (defanged)	Public IP lookup — anti-geofencing check
Mirai XOR keys: 0x22355627,	XOR keys	Mirai C2 domain obfuscation — decoded at

0x544f4e49	runtime
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6. Attack Source Attribution

Attacker source country attribution is based on correlator-processed data. The United States and China together account for the plurality of attack-correlated events, consistent with the concentration of cloud VPS infrastructure in those countries rather than indicating direct state-sponsored activity.

Rank	Country	Correlated Events
1	United States	6,092
2	China	5,973
3	Romania	2,857
4	Germany	2,398
5	Netherlands	2,222
6	Russia	1,903
7	Vietnam	1,449
8	India	774
9	Bulgaria	743
10	Hong Kong	679
11	South Korea	629
12	France	568
13	Taiwan	523
14	Indonesia	505
15	Singapore	503

Attacker infrastructure by autonomous system (ASN correlation):

Provider / ASN	Correlated Hits	Notes
DigitalOcean	1,835	Largest single attacker infrastructure source
OVH	436	European cloud/VPS provider
Amazon AWS	430	Commercial cloud — easily provisioned VPS
GCS (Global Connectivity)	248	

Alibaba Cloud	184	APAC cloud provider
VPSVAULT.HOST LTD	180	Budget VPS provider
M247 Europe	91	European transit/hosting
Aeza International	21	SANCTIONED — EU/US sanctions designation
Proton66 OOO	8	Bulletproof hosting — known criminal infrastructure

7. Abuse Reporting Pipeline

The automated abuse reporting pipeline generated 184 incident reports during April 2026, covering 184 unique malicious IP addresses across 135 providers and 108 distinct abuse contacts. Of these, 35 reports (19%) were submitted directly to provider abuse desks; the remaining 149 are archived in the submission queue pending rate-limit windows.

Provider	ASN	Reports Generated
Microsoft Corporation	AS8075	8
Google LLC	AS396982	5
Korea Telecom	AS4766	5
UCLLOUD HK	AS135377	5
DigitalOcean	AS14061	4
Censys Inc	AS398324	3
Tencent Cloud	AS45090	3

8. ICS/OT Honeypot — First Observations (PL Node)

An ICS/OT-focused honeypot node was launched on April 28, 2026, in Poland. The node emulates industrial control system protocols commonly deployed in energy sector and critical infrastructure environments. The geographic placement in Poland is deliberate, reflecting the heightened threat environment for industrial infrastructure in Eastern Europe and the documented interest from state-aligned threat actors in ICS/OT targeting across the NATO eastern flank.

Protocol	Standard	Typical Deployment
Modbus TCP	IEC 61158	SCADA, PLCs, industrial sensors
IEC 61850	IEC 61850	Substation automation, power sector
DNP3	IEEE 1815	Energy, water, oil & gas (US/Canada)
S7comm	Siemens S7	Siemens SIMATIC programmable controllers

With only three days of data available (April 28–30), meaningful statistical analysis is premature. Initial observations confirm that the node is receiving network traffic on monitored ICS ports, consistent with internet-wide scanning activity documented for ICS protocol ports (502/Modbus, 102/S7, 20000/DNP3, and associated IEC 61850 ports). Full analysis will be included in the May 2026 monthly report.

9. Indicators of Compromise (IOC Summary)

All domain and URL indicators are defanged per TLP:CLEAR handling requirements. Replace [.] with . and hxxps with https before use in security tools.

Type	Value	Context / Description
Domain (C2)	api[.]telegram[.]org	Telegram Bot API — malware C2; 3,383 TLS sessions observed
Domain (C2)	discord[.]com	Discord webhook C2 — Panchan / Discord_C2_ELF samples
Domain (C2)	discordapp[.]com	Discord API endpoint — same C2 cluster
Domain (miner)	xmrig[.]com	XMRIg cryptominer — distribution endpoint
Domain (pool)	nicehash[.]com	NiceHash mining pool — attacker payout endpoint
Domain (recon)	api[.]ipify[.]org	Public IP lookup — anti-geofencing technique
Network (ASN)	Aeza International (AS210644)	Sanctioned hosting provider; 21 attack-correlated hits
Network (ASN)	Proton66 OOO (AS198953)	Bulletproof hosting provider; 8 attack-correlated hits
YARA	WannaCry_Ransomware	884 PE32 sample matches — SMB/EternalBlue propagation
YARA	Panchan_SSHWorm_Go	6 samples — Go SSH worm with Discord C2
YARA	Discord_C2_ELF	7 ELF samples — Discord webhook exfil/C2
YARA	MAL_ARM_LNX_Mirai_Mar13_2022	2 ARM ELF samples — Mirai botnet variant
XOR Key	0x22355627	Mirai C2 domain XOR obfuscation key
XOR Key	0x544f4e49	Mirai C2 domain XOR obfuscation key

TLS SNI	api[.]telegram[.]org	3,383 sessions — malware C2 (DE + RU nodes)
Port	tcp/445	SMB — 80,500 connections (DE); WannaCry propagation vector
Port	tcp/11434	Ollama API — 54,032 unauthorized inference events
Port	udp+tcp/5060	SIP/VoIP — 236,573 events across all nodes
Username	sol / solana / solv	Solana validator SSH brute-force targets — 37,021 attempts
HASSH	af8223ac9914f509afdadfaf5f7ee94e	Top SSH client fingerprint — 28,005 sessions
HASSH	16443846184eafde36765c9bab2f4397	Second SSH client fingerprint — 24,171 sessions
HASSH	01ca35584ad5a1b66cf6a9846b5b2821	Third SSH client fingerprint — 21,527 sessions

10. MITRE ATT&CK Mapping

Observed attacker behaviors mapped to the MITRE ATT&CK framework (Enterprise and ICS matrices).

Technique ID	Name	Observed Evidence
T1110	Brute Force	578,657 SSH login attempts; targeted Solana validator credentials (sol/solana/solv: 37,021 attempts)
T1595	Active Scanning	236,573 SIP/VoIP scan events; 80,500 SMB/445 connections; 7,442 WinBox; 1,494 Kubelet probe events
T1595.002	Vulnerability Scanning	EternalBlue/MS17-010 exploitation via SMB; Ollama API exposure probing (port 11434)
T1059	Command and Scripting Interpreter	7 shell script malware samples dropped via SSH/Cowrie sessions
T1496	Resource Hijacking	54,032 Ollama API unauthorized inference events; Docker API llm-abuse (1,049 sessions); cryptominer samples (xmrig[.]com, nicehash[.]com)
T1071.001	Application Layer Protocol:	Malware C2 via HTTPS to api[.]telegram[.]org

	Web Protocols	(3,383 TLS sessions)
T1102	Web Service	Telegram Bot API as C2 (3,383 sessions); Discord webhook C2 (Panchan, Discord_C2_ELF — 7 samples)
T1583.003	Acquire Infrastructure: Virtual Private Server	Attacker infrastructure predominantly VPS: DigitalOcean (1,835 hits), OVH (436), AWS (430)
T1584	Compromise Infrastructure	Aeza International (sanctioned, 21 hits) and Proton66 OOO (bulletproof, 8 hits) used as attack sources
T1027	Obfuscated Files or Information	UPX-packed ELF samples (7 hits); Mirai XOR domain obfuscation (keys 0x22355627, 0x544f4e49)
T1078	Valid Accounts	30,984 successful honeypot logins; harvesting of SSH keys by Panchan worm for lateral movement
T1021.004	Remote Services: SSH	Panchan SSH worm propagation; SSH brute- force across all three primary nodes; 1,921 unique attacker IPs
T1498	Network Denial of Service	DDoSia/NoName057 monitoring: 18 C2 nodes (14 online), 153 active targets
T0840	Network Connection Enumeration (ICS)	ICS/OT protocol scanning on PL node (Modbus/502, S7/102, DNP3/20000, IEC 61850) — initial observations

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